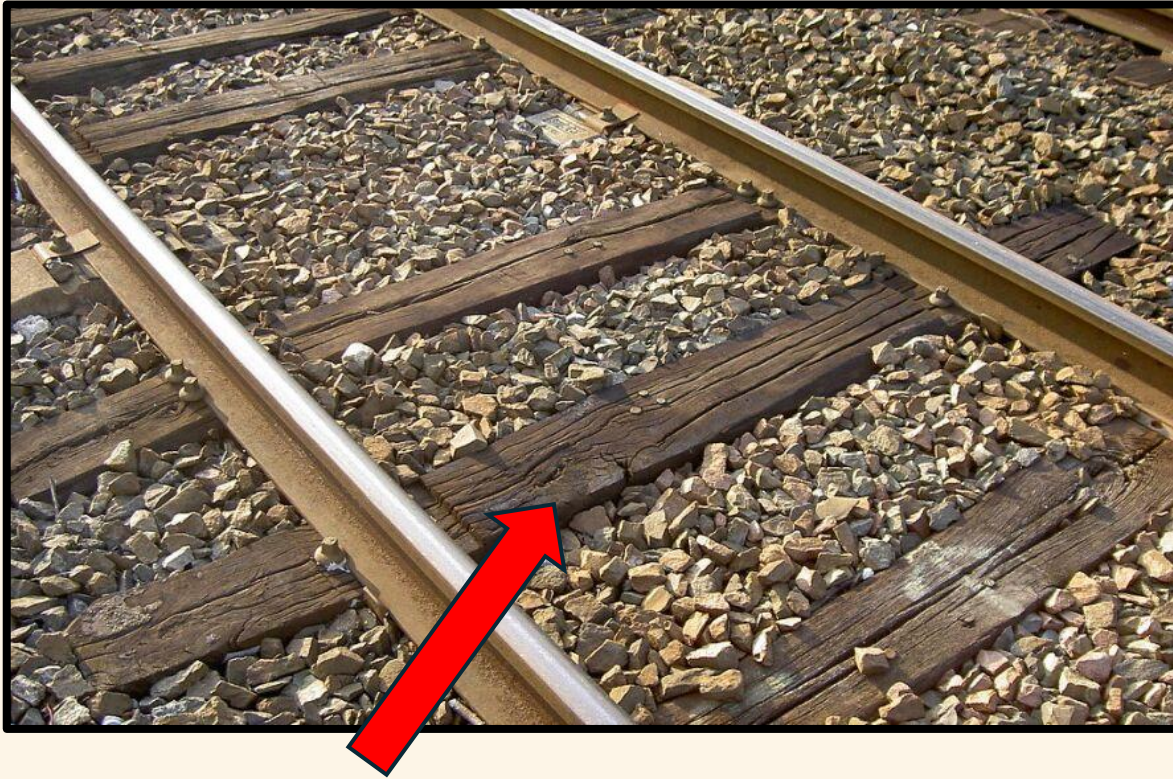


POLYRAIL



THE WOODEN RAILWAY TIE



1. Deforestation

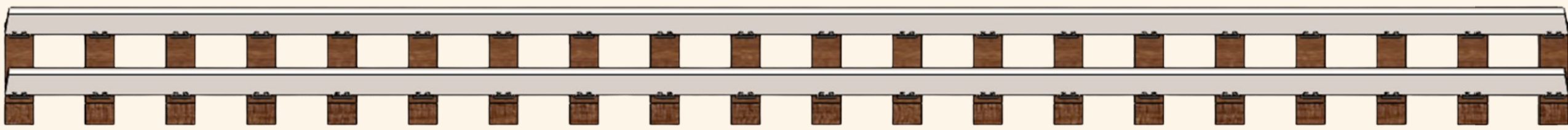
20 Million trees cut annually for U.S Rails. \$2B deforestation industry.

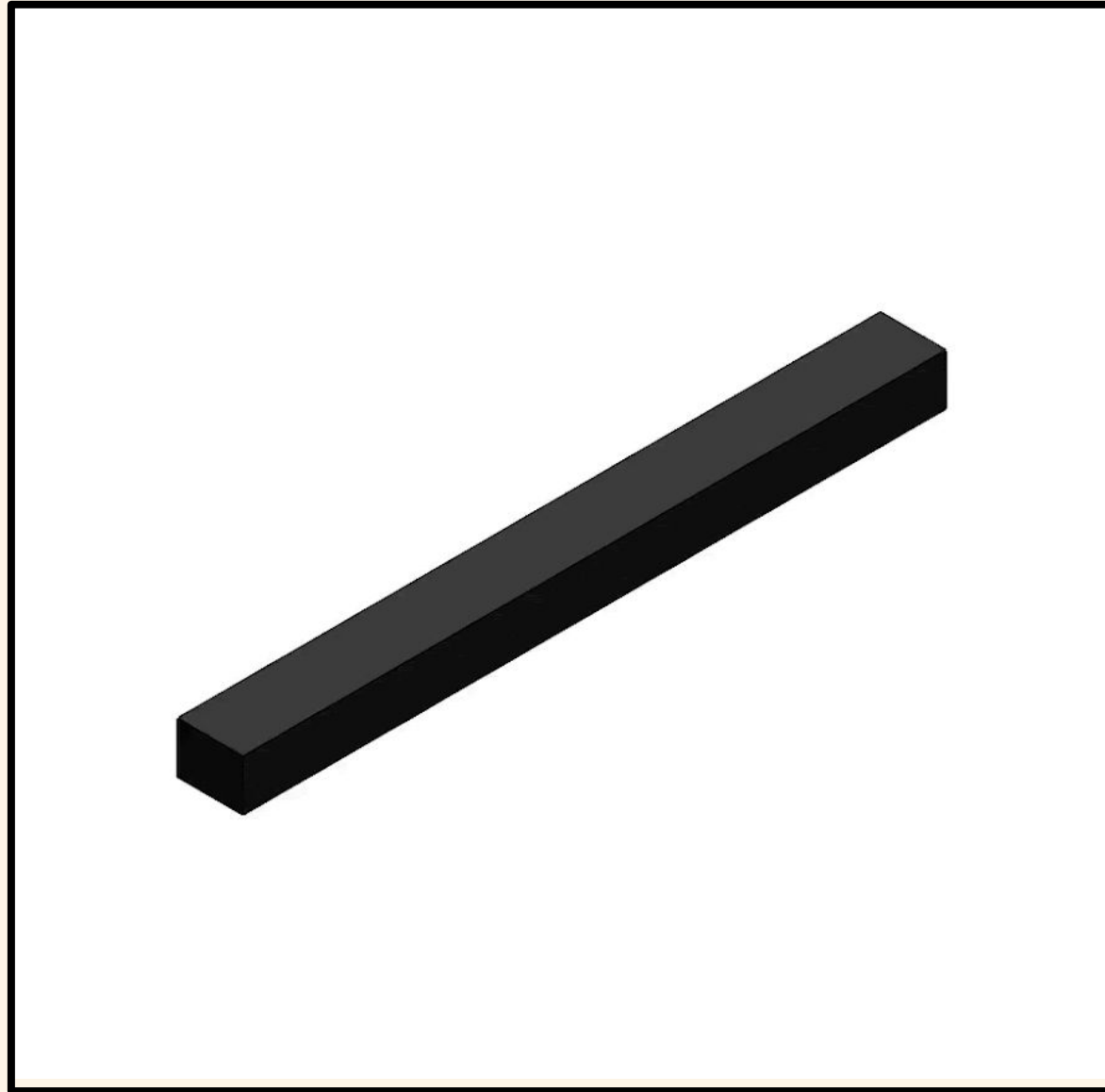
2. Wood rot

Ties rot and require replacement every 10 years

3. Cancer coating

Treated with creosote, a toxic preservative and carcinogen that seeps into soil and water supplies





THE POLYRAIL

THE ENVIRONMENTAL ALTERNATIVE

TITAN

Two birds, one stone

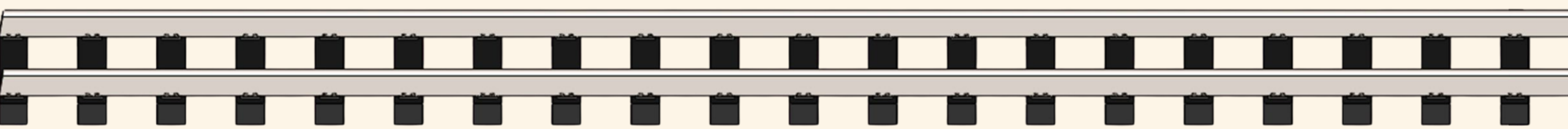
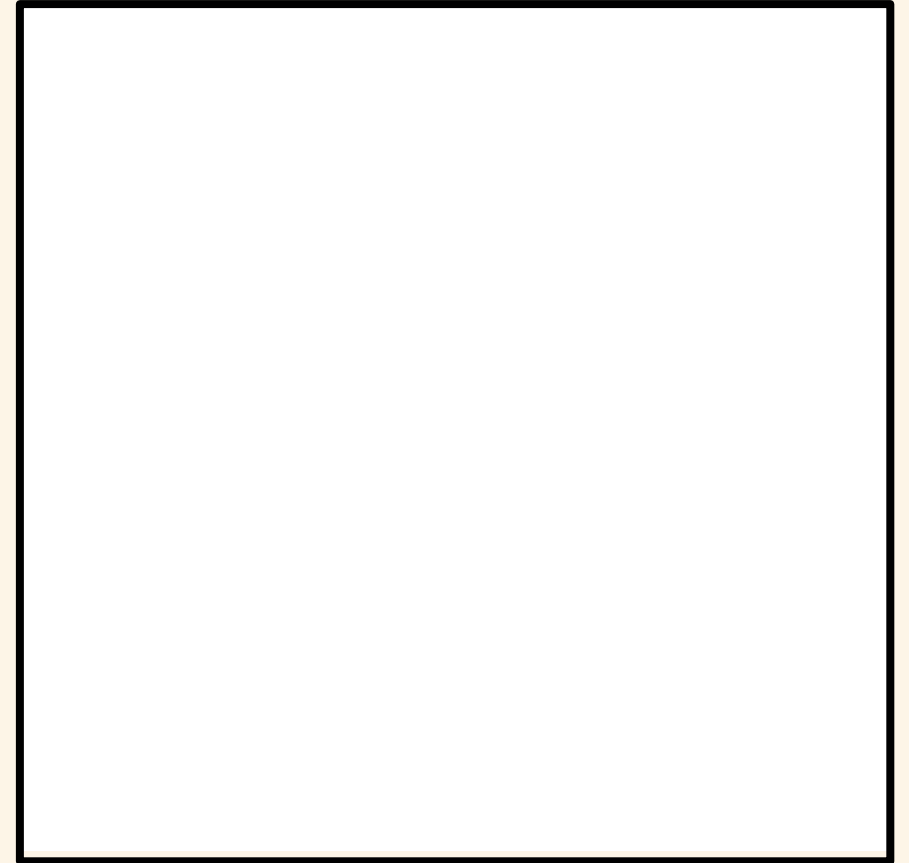
Polymer engineered from unrecycled #3,#6,#7 plastics, reducing both plastic waste and deforestation.

Durable

Lasts 50+ years (vs.10 for wood)

Cancer-free

No carcinogen chemicals seeping into soil and water supplies.



Agenda



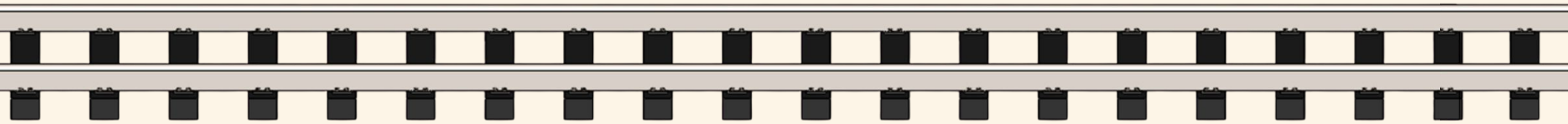
Design/Fabrication



Scalability/Marketing



Timeline/Projection



DESIGN/FABRICATION

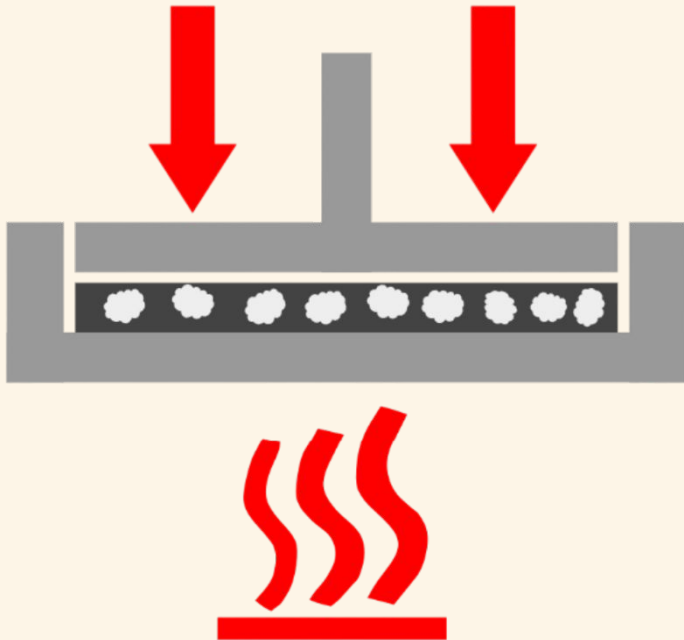
Plastic sintering

Polymer fused via pressure and heat

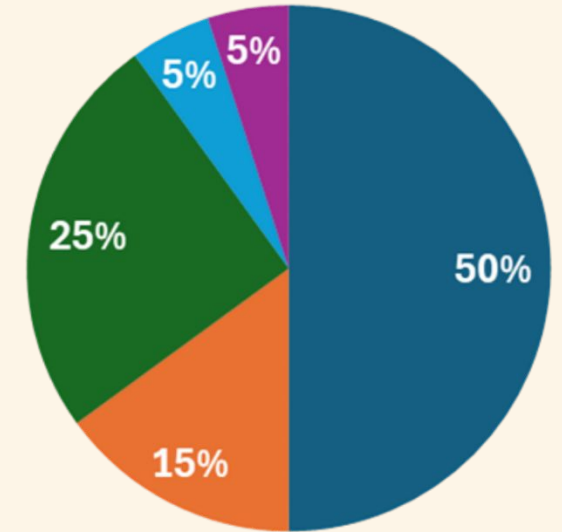
15% PVC:85% other

Meets stress requirements

Meets impact resistance requirements



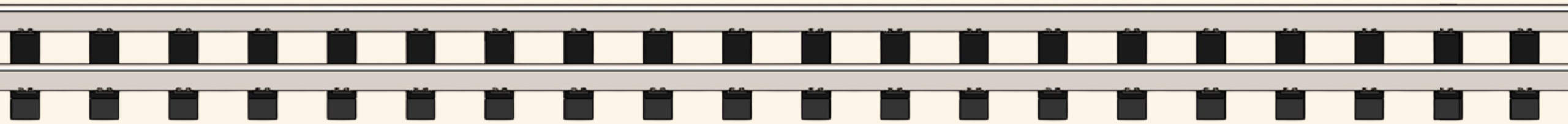
Percentage by Weight



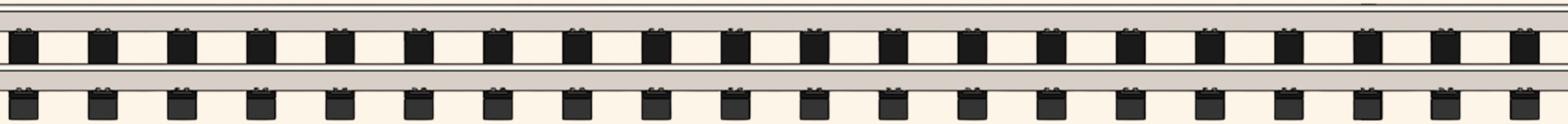
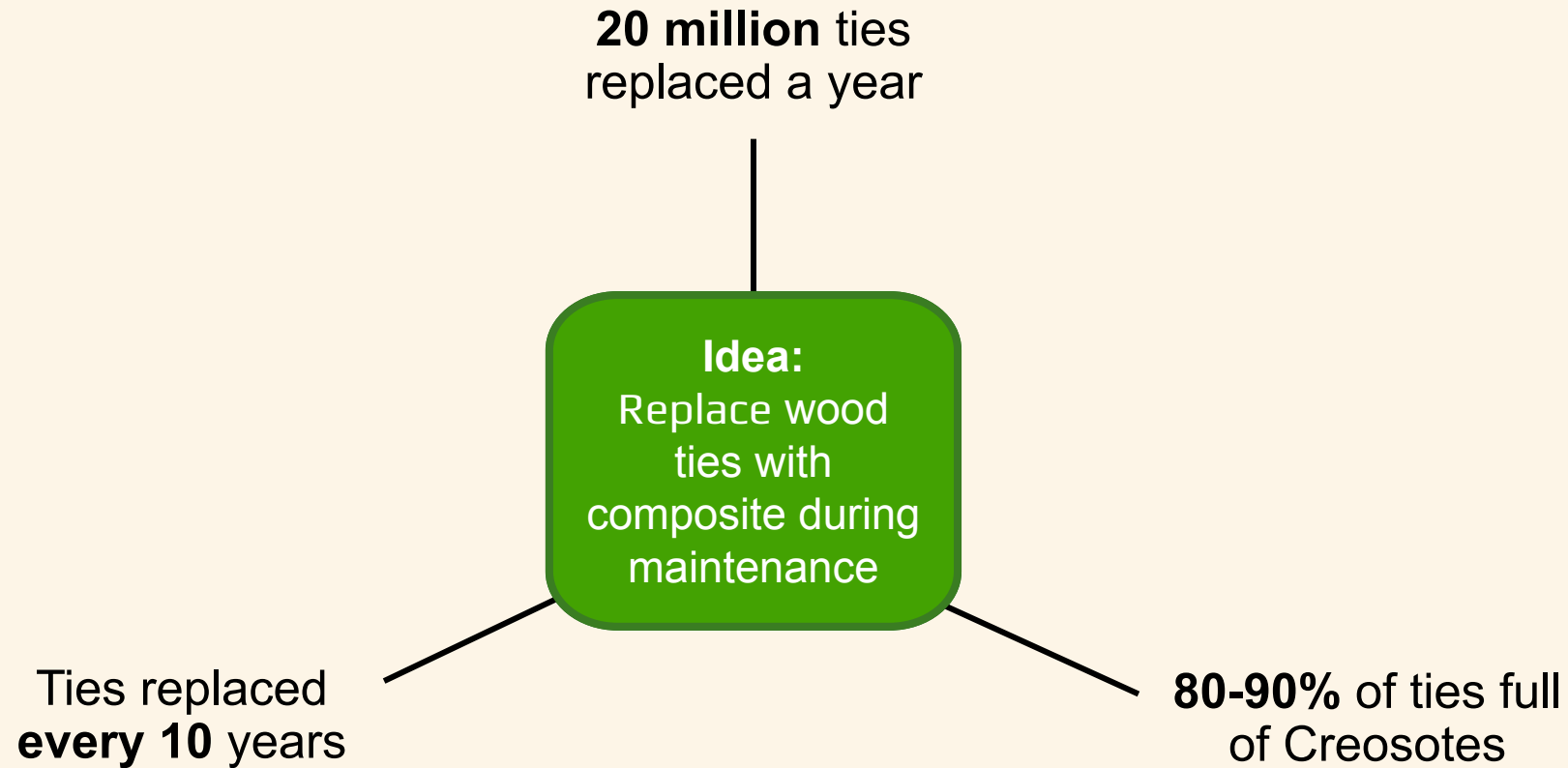
We take their plastic, they pay us

We partner with local Municipal recycling facilities, who source our composite materials

- Recycled polyolefin matrix (HDPE/PP)
- Recycled PVC and #7 plastics
- Glass fiber reinforcement
- Mineral and industrial waste fillers
- Additives/compatibilizers



Scalability



Target audience

Railway Contractors & Infrastructure Companies

3rd Party Logistics

Government & Municipal Agencies



BUILDING AMERICA®



PSC GROUP



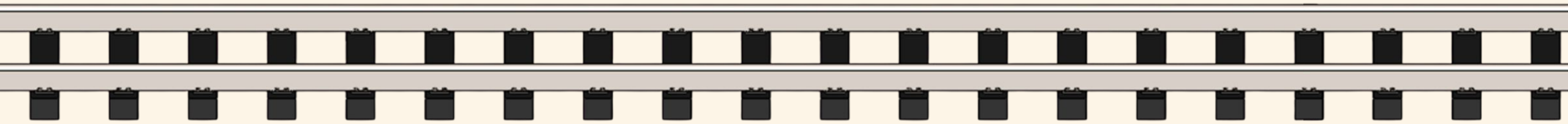
Why us?

	1 Wood Ties	1 PolyRail Titan Tie
Price	\$25	\$45
Replacement frequency	10 Years	50+ years
Cost over 50 years	\$250	\$45



Marketin

- **G** Industry Conferences
- Social media
- Sales Team



Timeline



Year 1

Foundation & validation

Research and development
Finalize composite formulation

Ties sold: 0

Year 3

Texas expansion

First Factory
Pitch to Union Pacific & expand within TX

Ties sold: 18k

Year 5

National expansion

Expand deployments to railroad corridors
Develop next-generation composite formulation

Ties sold: 300k

Year 2

Pilot & field testing

Expand capabilities for increased production
Keep within Bryan - College Station, TX

Ties sold: 1.8K

Year 4

Regional expansion

Scale production and logistics
Expand within Southern US

Ties sold: 90k

Projections



Costs

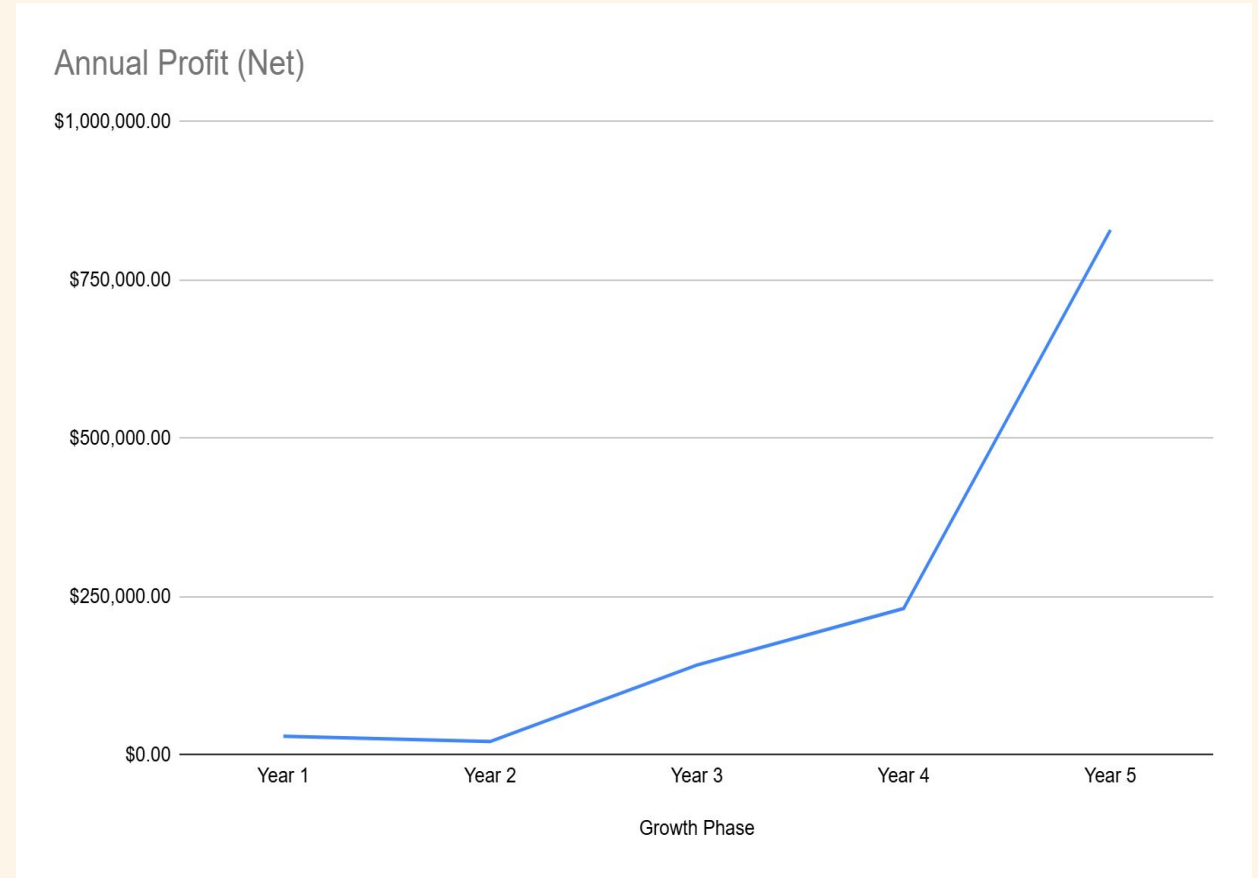
Raw Materials, Logistics, Labor,
Operations, Legal/Insurance

Revenue

Selling ties, plastic buyback program

Grants

Aggie Pitch, EPA P2 Grant, College
Station EDC Micro Grant



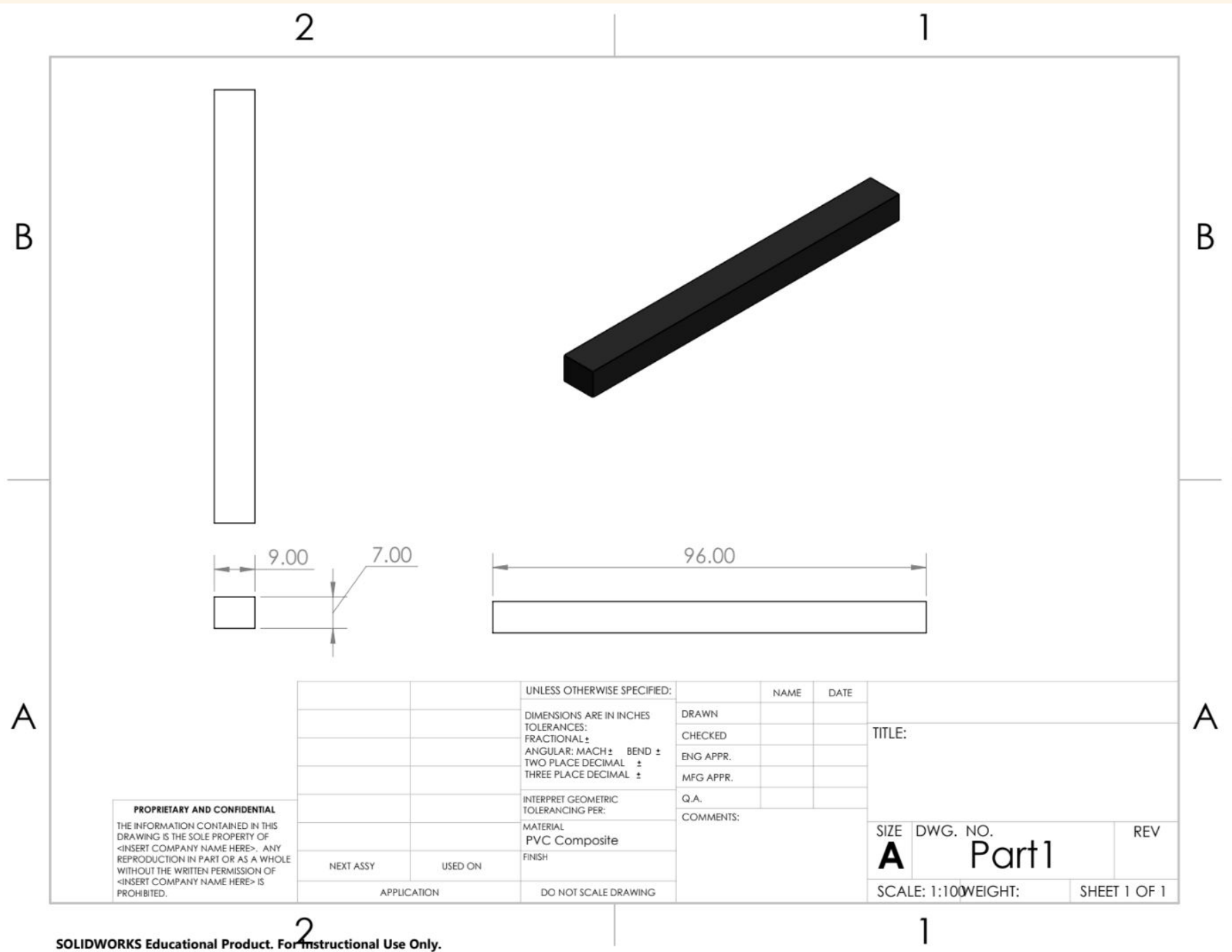
Future aspirations

Growth into
international markets.

Reaching the
undesirable plastics
in countries with
lacking infrastructure
to properly recycle.

Expanding our scopes
into other railway
systems beyond the
US borders, pushing
our goal of impacting
the
environment further.





Cost

Item	Wood Ties	PolyRail Titan Ties
Upfront cost per tie	-25 \$	45 \$
Service life (years)	10	50+
Upfront cost per mile	83,750 \$	150,750 \$
Cost of 1 tie per year	2.5 \$/yr	0.9 \$/yr
Cost of 1 mile per year	8,375 \$/yr	3,015 \$/yr
50-year lifecycle cost per mile	418,750 \$ (5× replacement)	150,750 \$ (1 install)
Total savings over 50 years		268,000 \$ saved per mile

Year 1 Finances

	Number to Titan Ties produced Monthly	5		Number of Titan Ties Sold Monthly	0	
	Workers on Payroll	0				
	Cost Type	Item Description	Price/Unit	Price/Month	Price/Year	Notes / Realistic Adjustments
Sources						
UNIT COSTS			Price/tie			
https://www.cher	Raw Materials	Glass Fiber (30%)	\$ (17.59)	\$ (87.95)	\$ (1,055.40)	30% Mix @ \$0.27/lb
https://www.cher	Raw Materials	Recycled HDPE (55%)	\$ (8.60)	\$ (43.00)	\$ (516.00)	55% Mix @ \$0.08/lb
https://fmcpcapc	Raw Materials	PVC/#7 Waste (15%)	\$ 2.00	\$ 10.00	\$ 120.00	Income Offset (Tipping Fee to get plastic off their hands)
https://www.mad	Raw Materials	Additives/Stabilizers(5%)	\$ (7.82)	\$ (39.10)	\$ (469.20)	UV/Heat Stabilizers
https://redstagful	Logistics	Outbound Freight	\$ (8.00)	\$ (40.00)	\$ (480.00)	Shipping 150 heavy units
https://tees.tamu	Labor	TAMU interns/jobs	\$ (5,000.00)	\$ -	\$ -	\$15/hr @ 20hrs/week
MONTHLY COSTS						
https://www.getr	Operations	Manufacturing Scrap (5%)		\$ (450.00)	\$ (5,400.00)	Buffer for failed units
https://www.uline	Operations	Maintenance & PPE		\$ (350.00)	\$ (4,200.00)	Machine wear & safety gear
https://btutilities.com	Utilities	Industrial Electricity		\$ (400.00)	\$ (4,800.00)	High-voltage for machines
https://www.mon	Legal/Insurance	Liability & Workers Comp		\$ (450.00)	\$ (5,400.00)	Higher rate for manufacturing
	Equipment	Compressor/ Plastic Densifier Rental		\$ (450.00)	\$ (5,400.00)	
STARTUP COSTS						
	Certification	AREMA Safety Testing			\$ (20,000.00)	Official Railroad Standard
	Testing	ASTM Lab Bench Tests			\$ (3,500.00)	Pre-certification strength proof
	R&D	Prototype Run			\$ (2,000.00)	First 50 test units
	Patent	Patent Filing			\$ (3,500.00)	Provisional Patent
	Business Dev	Travel & Pitching			\$ (5,000.00)	Finding Class 1 partners
MONTHLY REVENUE						
		Titan Ties Sales	\$ 45.00	\$ -	\$ -	
		Tipping Fees (Waste) MSW		\$ 100.00	\$ 1,200.00	
STARTUP FUNDING						
		EPA P2 Grant			\$ 55,000.00	
		Aggie Pitch			\$ 10,000.00	
		College Station EDC Micro Grant			\$ 25,000.00	
	YEAR TOTAL				\$ 29,599.40	
	LIFETIME TOTAL				\$ 29,599.40	

Year 2 Finances

	Number to Titan Ties produced Monthly	150		Number of Titan Ties Sold Monthly	150	
	Workers on Payroll	1				
	Cost Type	Item Description	Price/Unit	Price/Month	Price/Year	Notes / Realistic Adjustments
Sources						
UNIT COSTS			Price/tie			
https://www.cheerup.com/	Raw Materials	Glass Fiber (30%)	\$ (17.59)	\$ (2,638.50)	\$ (31,662.00)	30% Mix @ \$0.27/lb
https://www.cheerup.com/	Raw Materials	Recycled HDPE (55%)	\$ (8.60)	\$ (1,290.00)	\$ (15,480.00)	55% Mix @ \$0.08/lb
https://fmcpcap.com/	Raw Materials	PVC/#7 Waste (15%)	\$ 2.00	\$ 300.00	\$ 3,600.00	Income Offset (Tipping Fee to get plastic off their hands)
https://www.madsci.org/	Raw Materials	Additives/Stabilizers(5%)	\$ (7.82)	\$ (1,173.00)	\$ (14,076.00)	UV/Heat Stabilizers
https://redstagful.com/	Logistics	Outbound Freight	\$ (8.00)	\$ (1,200.00)	\$ (14,400.00)	Shipping 150 heavy units
https://tees.tamu.edu/	Labor	TAMU interns/jobs	\$ (2,500.00)	\$ (2,500.00)	\$ (30,000.00)	\$15/hr @ 20hrs/week
MONTHLY COSTS						
https://www.gettyimages.com/	Operations	Manufacturing Scrap (5%)		\$ (450.00)	\$ (5,400.00)	Buffer for failed units
https://www.uline.com/	Operations	Maintenance & PPE		\$ (350.00)	\$ (4,200.00)	Machine wear & safety gear
https://btutilities.com/	Utilities	Industrial Electricity		\$ (400.00)	\$ (4,800.00)	High-voltage for machines
https://www.monster.com/	Legal/Insurance	Liability & Workers Comp		\$ (450.00)	\$ (5,400.00)	Higher rate for manufacturing
https://www.svb.com/	Marketing	Web & Outreach		\$ (100.00)	\$1,200.00	Digital presence/ ads, etc
	Equipment	Compressor/ Plastic Densifier Rental		\$ (450.00)	\$ (5,400.00)	
MONTHLY REVENUE						
		Sales (0 Titan Ties) per month	\$ 45.00	\$ 6,750.00	\$ 81,000.00	
		Tipping Fees (Waste) MSW		\$ 100.00	\$ 1,200.00	
STARTUP FUNDING						
		EPA P2 Grant		\$	55,000.00	
		Aggie Pitch		\$	10,000.00	
	YEAR TOTAL			\$	21,182.00	
	LIFETIME TOTAL			\$	50,781.40	

Total Annual Revenue vs. Total Annual Cost

